

COMPASS TEMPEST Sulfide

June 2026 Samples - 141 and 141A

2026-07-29

Code Set up

run information

```
##Before or After Reruns (YES or NO)
After_Reruns = "YES"
#If NO will not write final data file
#And Will Write a Rerun File

###things that need to be changed
sample_month = "June"
sample_year = "2026"
Run_Date = as.Date("07/24/2026", format = "%m/%d/%Y")
Run_by = "Zoe Read, Melanie Giessner" #Instrument user
Script_run_by = "Zoe Read" #Code user

Run_notes="All dups and related samples had values of 0.
11/16 spikes were good." #any notes from run

#Plate Data
folder_path_rawdata <- paste0("Raw Data")
Raw_plates <- list.files(path = folder_path_rawdata, full.names = TRUE, pattern = "20260724")
(Raw_plates)

## [1] "Raw Data/COMPASS_H2S_20260724_Plate4_tidy.csv"
## [2] "Raw Data/COMPASS_H2S_20260724_Plate5_tidy.csv"
## [3] "Raw Data/COMPASS_H2S_20260724_Plate6_tidy.csv"

#Sample ID Path
SampleID_path = "Raw Data/TEMPEST_S04_20260501-20260612_corrected.csv"

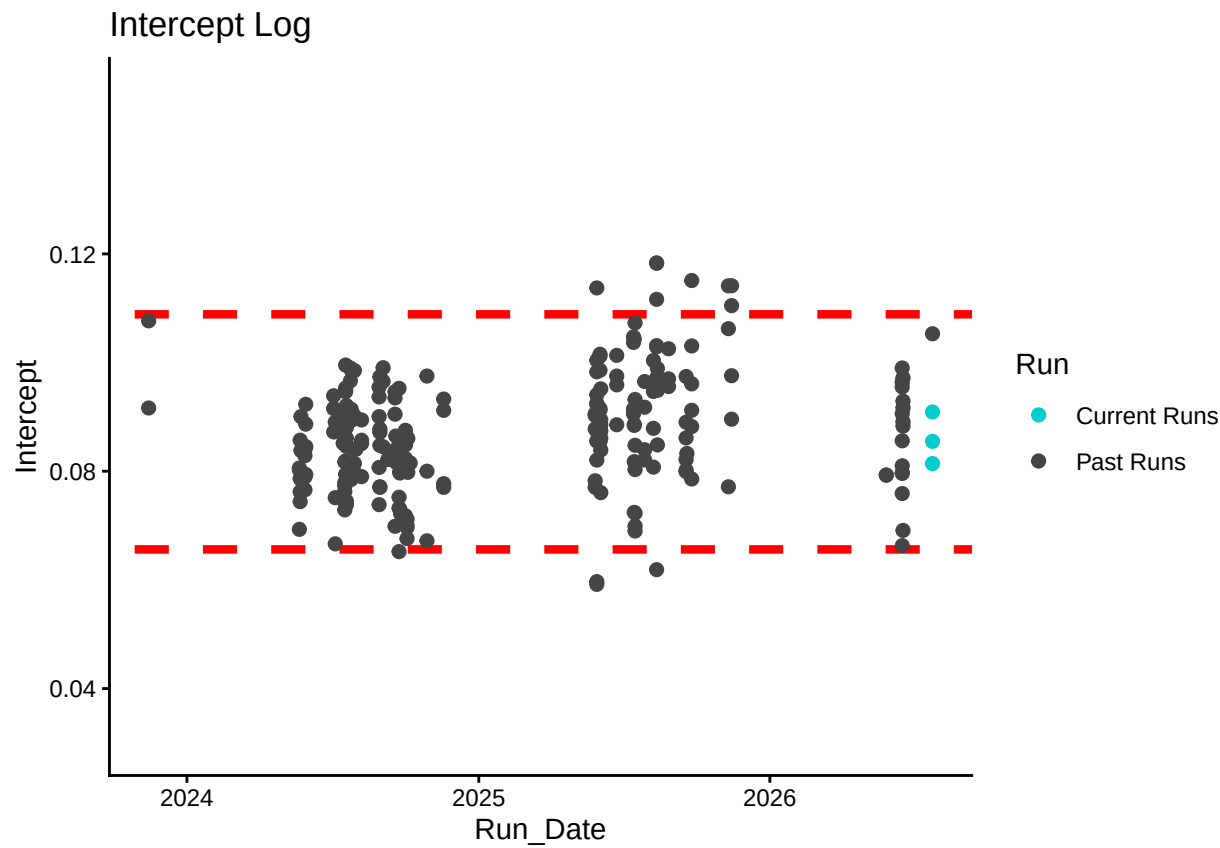
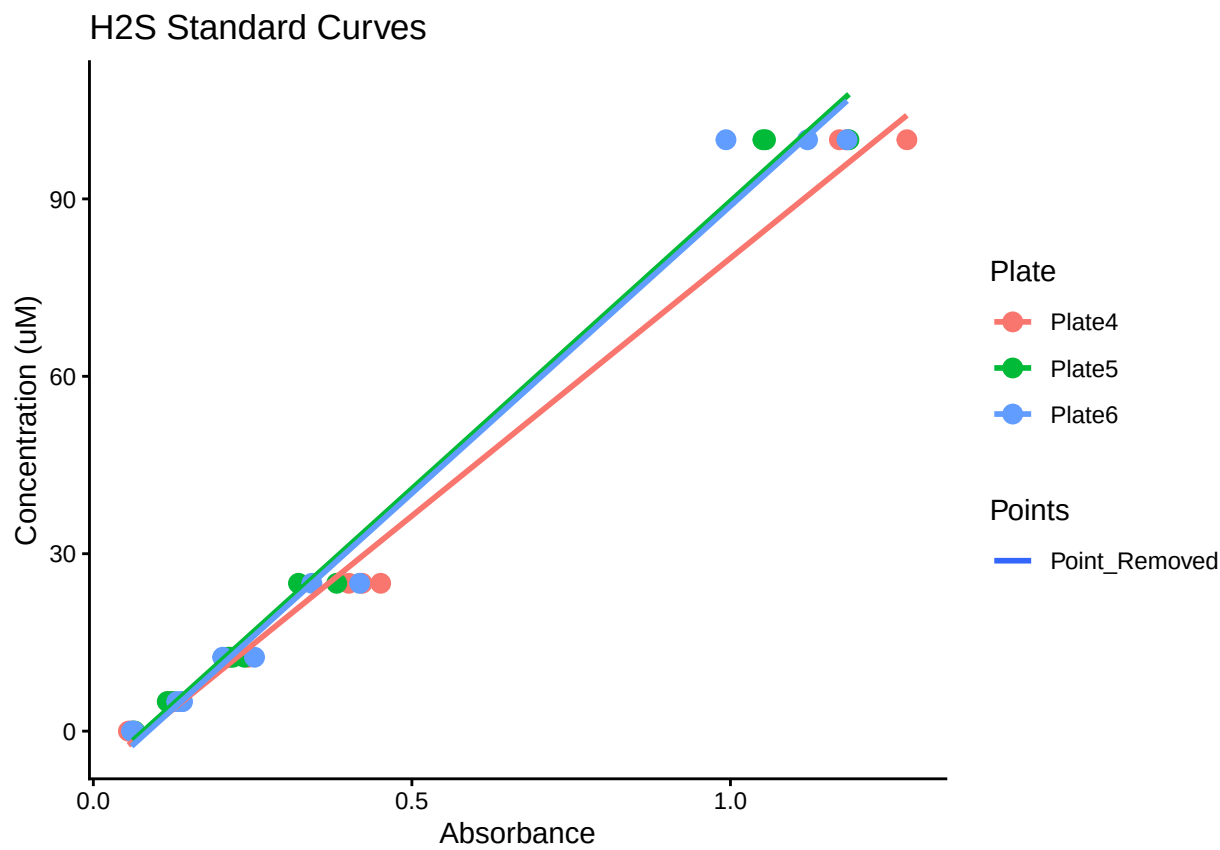
# Define the file path for QAQC log file - NO Need to change just check year
log_path <- "Raw Data/Sulfide_STD_QAQC.csv"

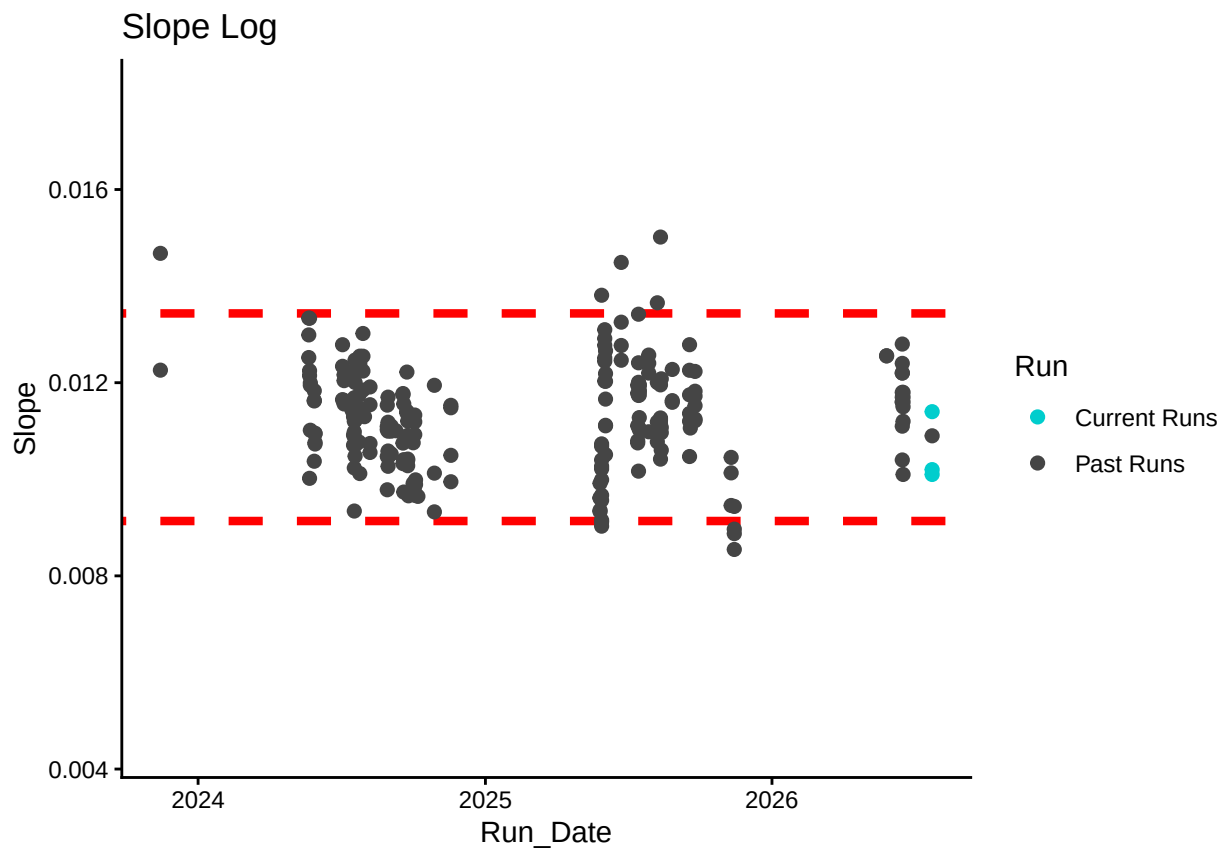
# Define final path for data be sure to change year and month and run number
final_path <- "Processed Data/COMPASS_TEMPEST_H2S_202606_141_141A_processed.csv"
rerun_path <- "Processed Data/COMPASS_TEMPEST_H2S_202606_reruns.csv"

##Read in data
##Remove High CV Stds
##Check R2 & Remove Standards if Necessary

## Joining with 'by = join_by(Plate, IDs)'
```

Plot Standard Curves



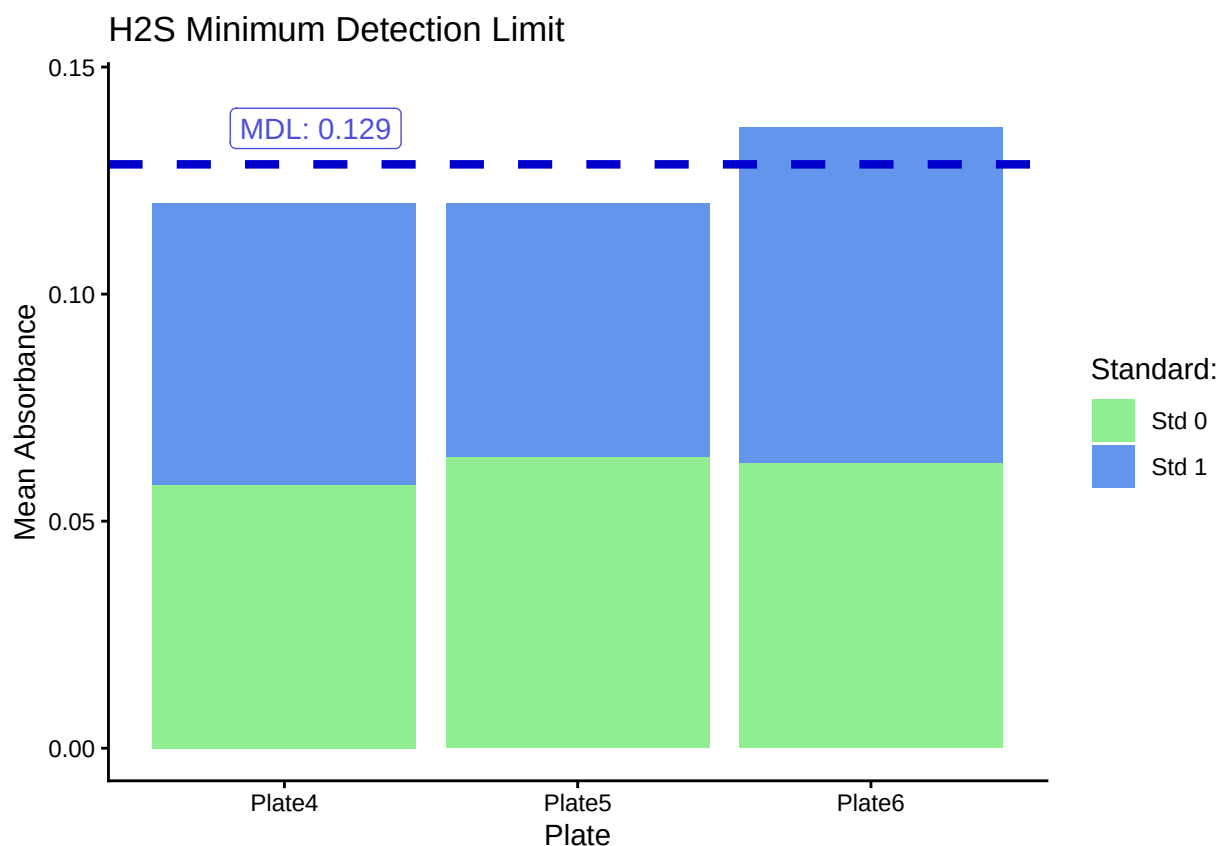


Return Flags For Standard Curve

Plate	R2_Flag	CV_Flag
Plate4	R2_Good	CV_Good
Plate5	R2_Good	CV_Good
Plate5	R2_Good	CV_High
Plate6	R2_Bad	CV_Good
Plate6	R2_Bad	CV_High

Plate	Curve_RMVD
Plate4	KEPT
Plate5	KEPT
Plate6	REMOVED

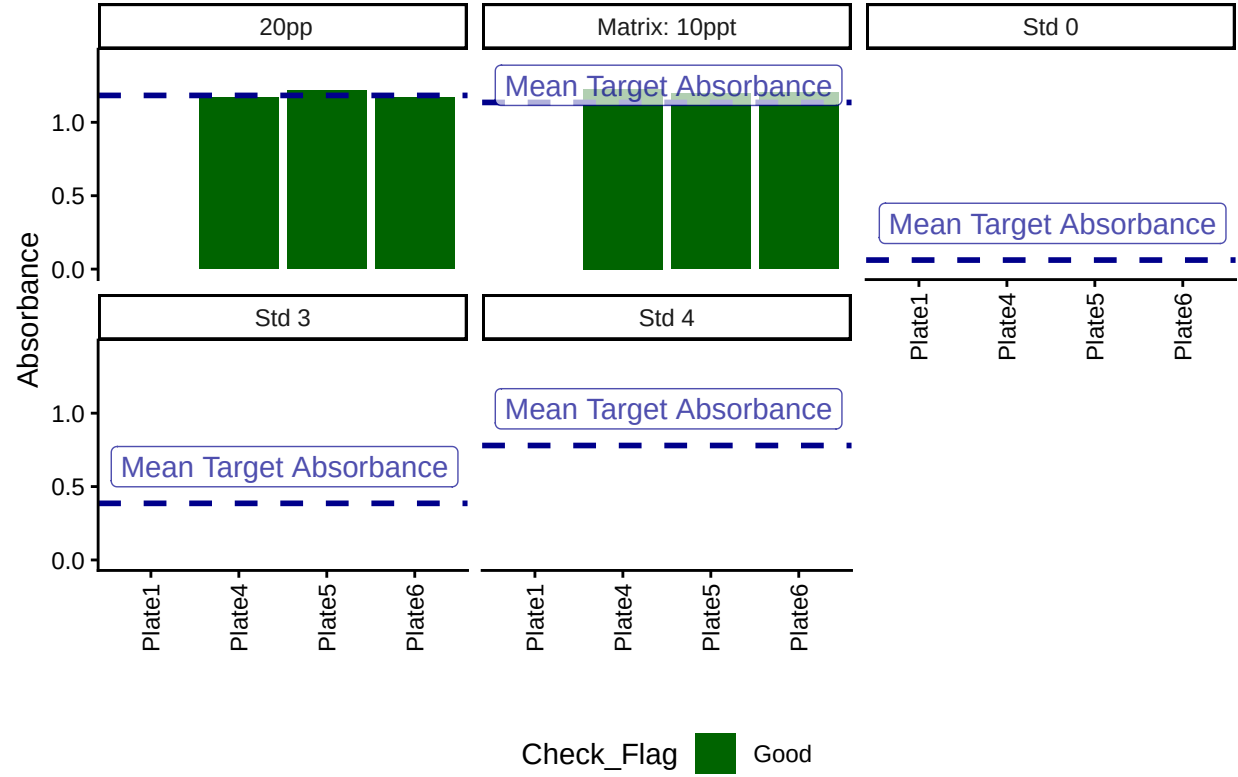
Method Detection Limit



Compare Check Standards to Standards

```
## Warning in min(x): no non-missing arguments to min; returning Inf
## Warning in max(x): no non-missing arguments to max; returning -Inf
## Warning in min(x): no non-missing arguments to min; returning Inf
## Warning in max(x): no non-missing arguments to max; returning -Inf
## Warning in min(x): no non-missing arguments to min; returning Inf
## Warning in max(x): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning in FUN(X[[i]], ...): no non-missing arguments to max; returning -Inf
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_label_repel()').
```

Check Stds: H2S with Confidence Interval



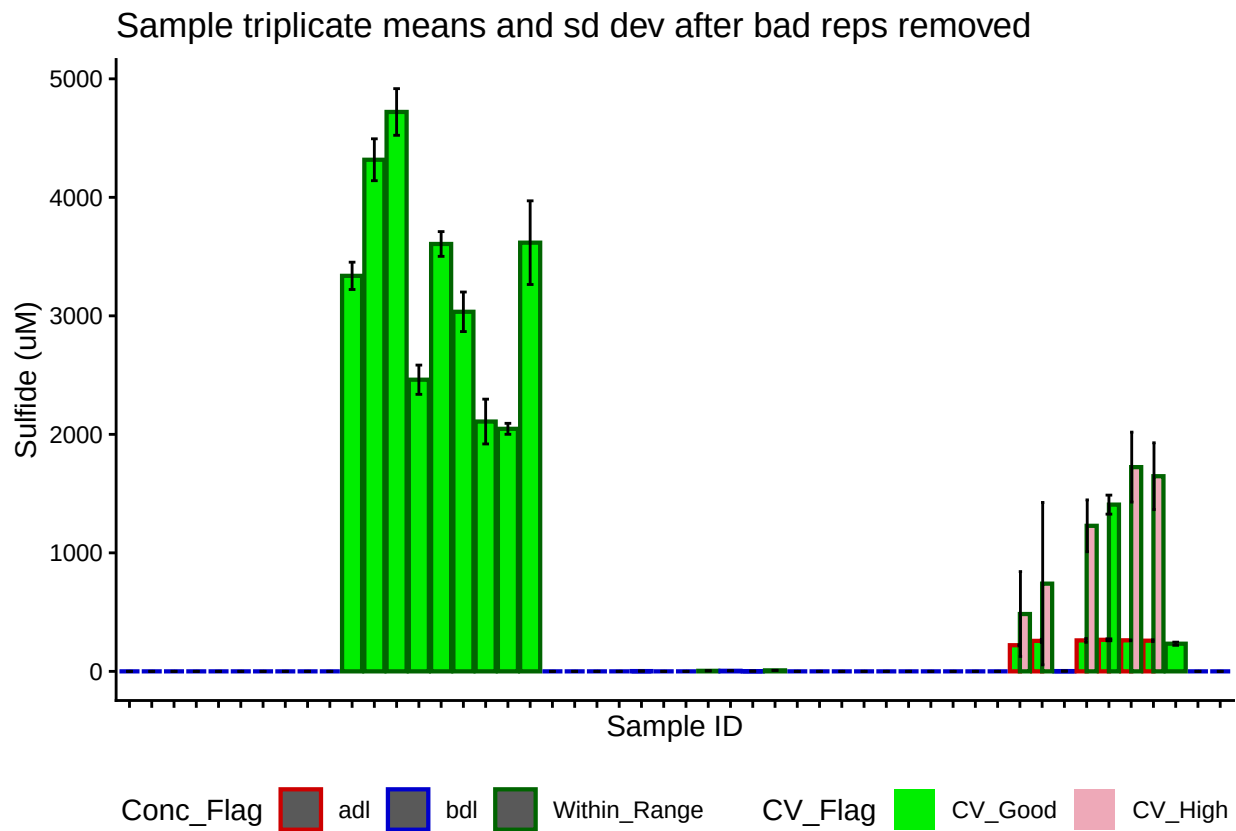
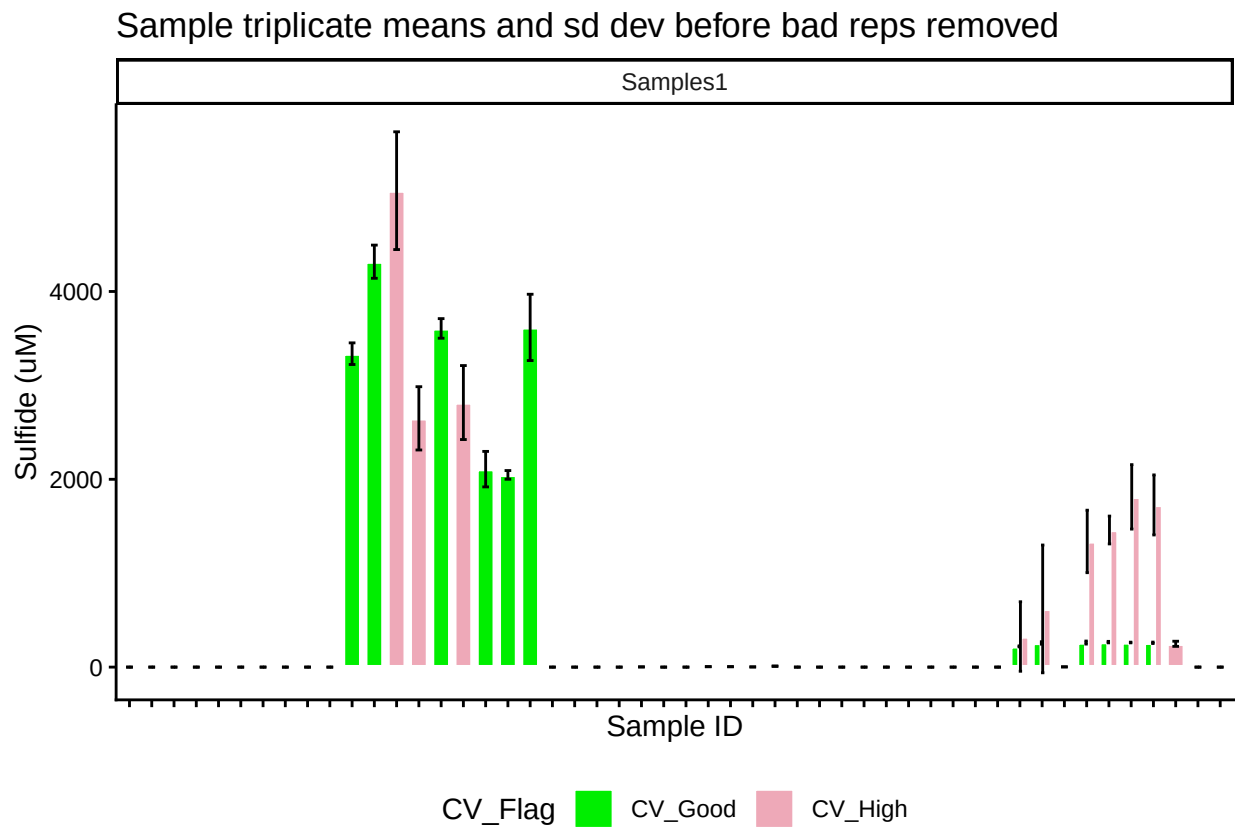
Display Any Check Flags

Plate	CHKs_Flag
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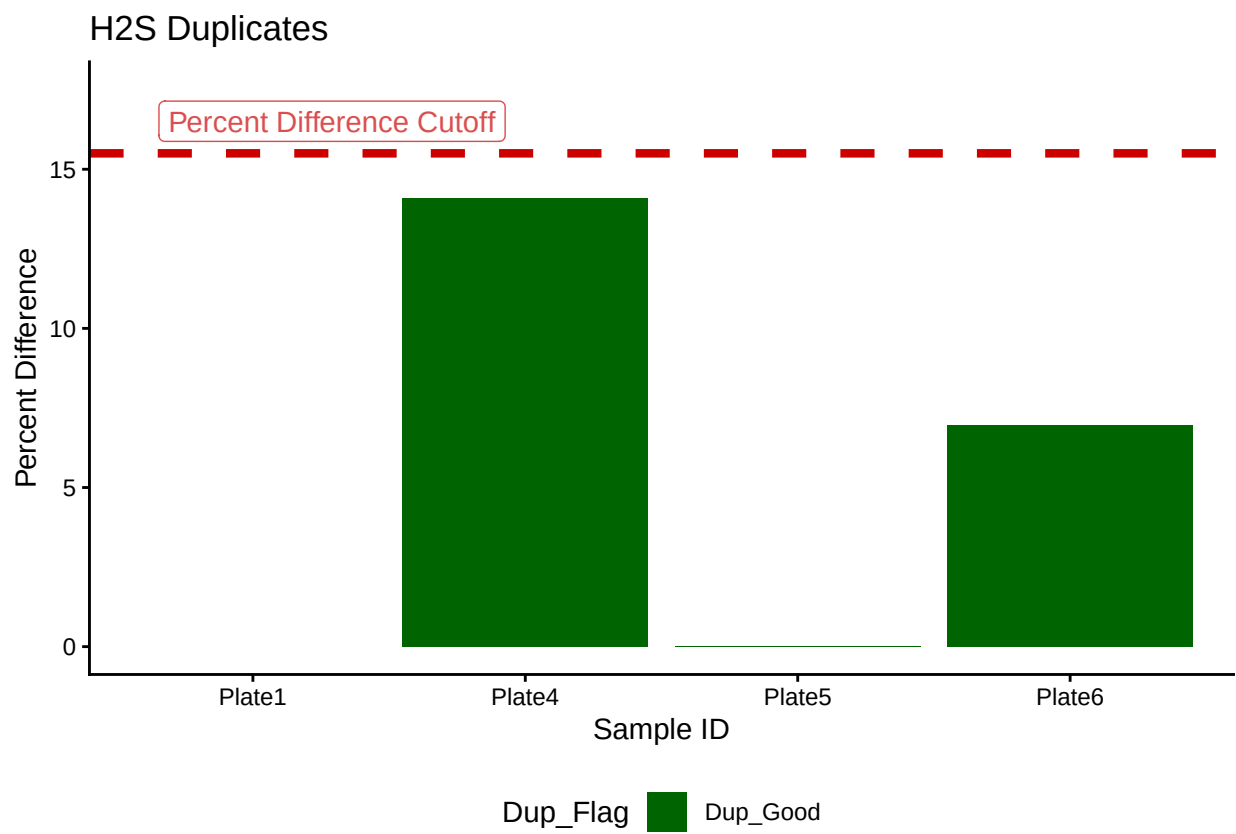
Plate	CHK_CV_Flags
-------	--------------

```
##Calculate Sulfide Concentrations & Add Concentration Flags
##Remove High CV Samples and Average
```

Plot Samples



Duplicate QAQC



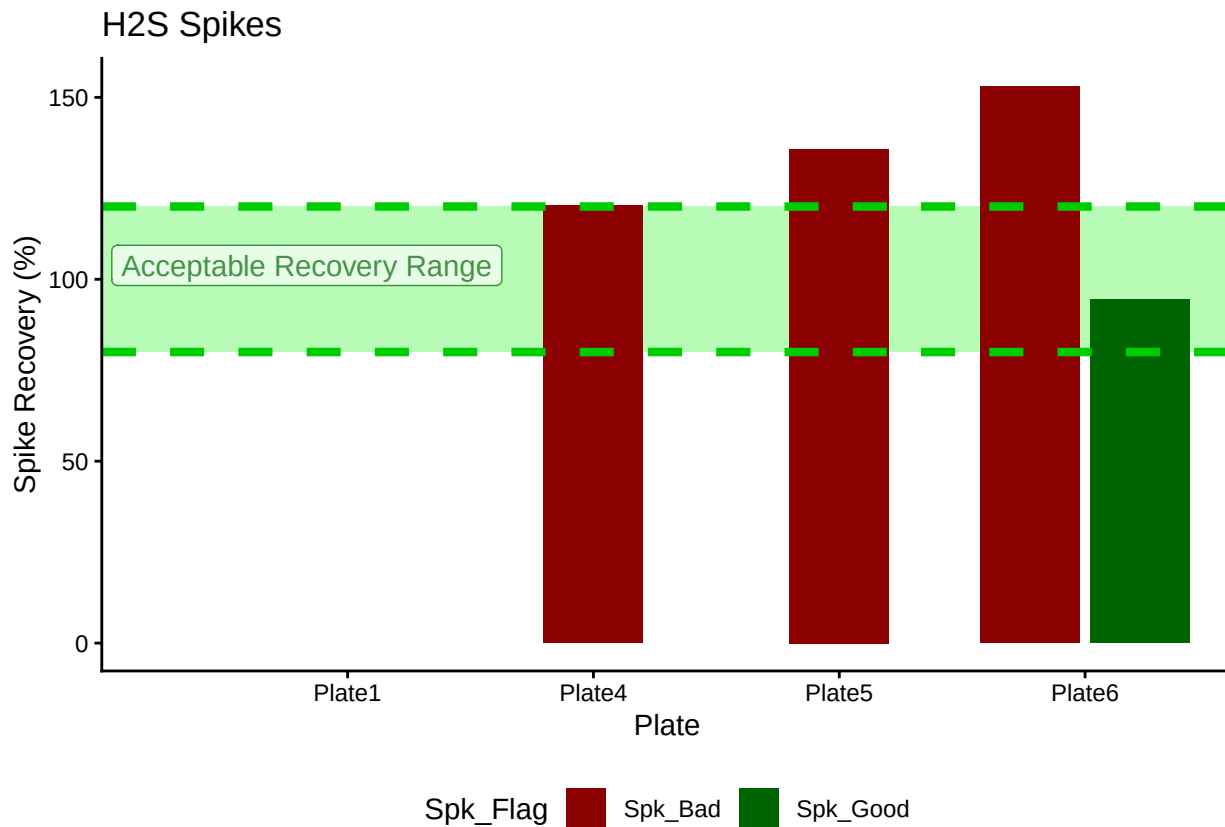
Display Dup Flags

```
## [1] ">60% Dups Pass"
```

Plate	Dup_Flag
Plate4	Dup_Good
Plate5	Dup_Good
Plate6	Dup_Good

Spike QAQC

```
## Warning in left_join(., spk_flags_1, by = join_by(Spk_ID, Plate)): Detected an unexpected many-to-many :
## i Row 6 of 'x' matches multiple rows in 'y'.
## i Row 1 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many" to silence this warning.
```



Display Spike Flags

```
## [1] "Spikes Paired"
```

Plate	Spike_Flag
Plate4	Spk_Bad
Plate5	Spk_Bad
Plate6	Spk_Bad

```
## [1] "<60% of Spikes Pass, Rerun"
```

Check for reruns and replace

```
##Add Flags
```

```
##Remove Synoptic samples and rename TEMPEST samples
```

```
df_all_clean <- df_all_clean1 %>%
  filter(!str_detect(Sample_ID, "SWH|GCW|GWI|MSM")) %>%
  filter(Sample_ID != "141") %>%
  mutate(Sample_ID = gsub("TMP_2026_EVENT_141A", "141A", Sample_ID)) %>%
  mutate(Sample_ID = gsub("TMP_2026_EVENT_141", "141", Sample_ID))
```

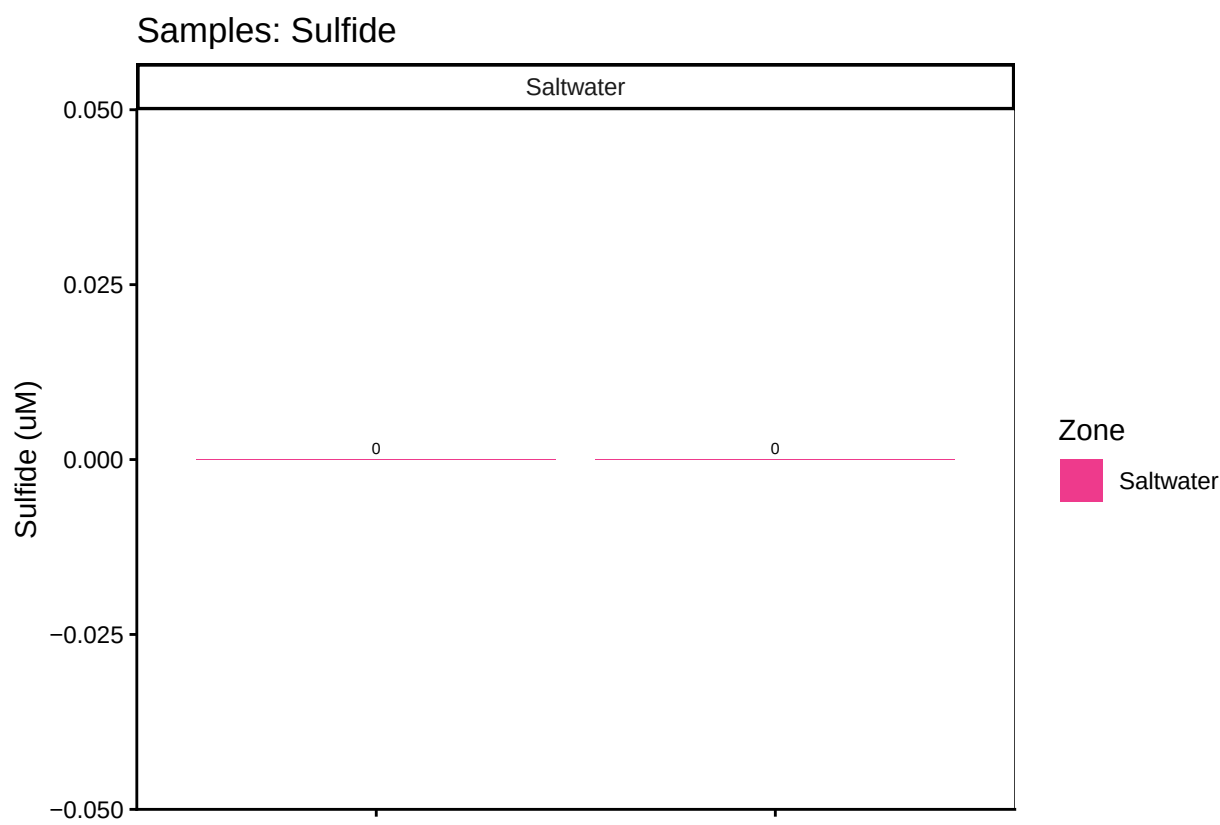
```
##Merge with Sample ID list to get actual sample IDs
```

```
##Pull in active porewater inventory
```

```
##Check samples against metadata
```

```
## ***All sample IDs are present in metadata.***
```


Visualize Data by Plot



##Export Data

END